Smart Structures: Piezoelectric Shunt Damping

Matlab Session Report: Modeling of a Tuned Mass Damper and its Equivalence to a Resonant Piezoelectric Shunt

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Abstract

This report details the modeling, optimization, and analysis of a Tuned Mass Damper (TMD) and its electromechanical equivalent, the piezoelectric resonant shunt. First, a 2-DOF mechanical system is analyzed and optimized, comparing numerical `fmincon` results with the analytical formulas of Liu & Liu (2005). Both methods converge to a near-identical optimal solution. Second, a direct electromechanical analogy is established between the mechanical TMD and a piezoelectric system with an R-L shunt circuit. This analogy is used to tune the piezoelectric shunt, and the results are compared against direct electrical tuning formulas. Finally, a detailed sensitivity analysis is performed to investigate the system's robustness to variations in shunt parameters, confirming the distinct roles of inductance (tuning) and resistance (damping).

Contents

1 Summary of Key Findings

This report details the modeling, optimization, and analysis of a Tuned Mass Damper (TMD) and its electromechanical equivalent, the piezoelectric resonant shunt.

1. **TMD Optimization:** A numerical optimization (`fmincon`) and an analytical formula (Liu & Liu, 2005) were both successfully used to design a TMD. Both methods converged to nearly identical parameters, validating the analytical model. The optimized TMD effectively suppressed the primary resonance peak, replacing it with two smaller, well-damped peaks, drastically reducing the maximum vibration amplitude.

2. **Electromechanical Analogy:** A direct analogy was established between the 2-DOF (Degree of Freedom) mechanical TMD system and a 1-DOF piezoelectric system with an R-L shunt circuit. The absorber's mass (m_a) and damper (c_a) were shown to be directly analogous to the shunt's inductance (L) and resistance (R), respectively.

3. **Piezoelectric Shunt Tuning:** The piezoelectric shunt was tuned using two different methods: (1) translating the optimal *mechanical* parameters from the Liu & Liu analogy and (2) using *direct* tuning formulas for piezoelectric shunts (Thomas et al., 2011). Both methods yielded effective damping.

4. **Sensitivity Analysis:** The system's robustness was confirmed. The analysis of extreme cases (L/R $\rightarrow$ 0 or ∞) correctly replicated a simple resistive shunt, an undamped resonant shunt, and an open-circuit condition. A $\pm$20% sensitivity analysis showed that while mistuning degrades performance (creating uneven peaks), the system is reasonably robust to small component variations.

2 Dynamics of a Double Mass System

This section establishes the baseline behavior of the un-optimized 2-DOF mechanical system, which consists of a primary mass (m) and an absorber mass (m_a).

2.1 Modal Analysis

An eigenvalue analysis was performed on the undamped, coupled system to find its fundamental modes of vibration. The two natural frequencies of the coupled system were calculated to be:

- $\omega_1 = 84.03$ rad/s
- $\omega_2 = 144.93$ rad/s

2.2 Time-Domain Response (Sine Input)

The system was simulated with a sinusoidal excitation force at $\omega_0 = 100$ rad/s.

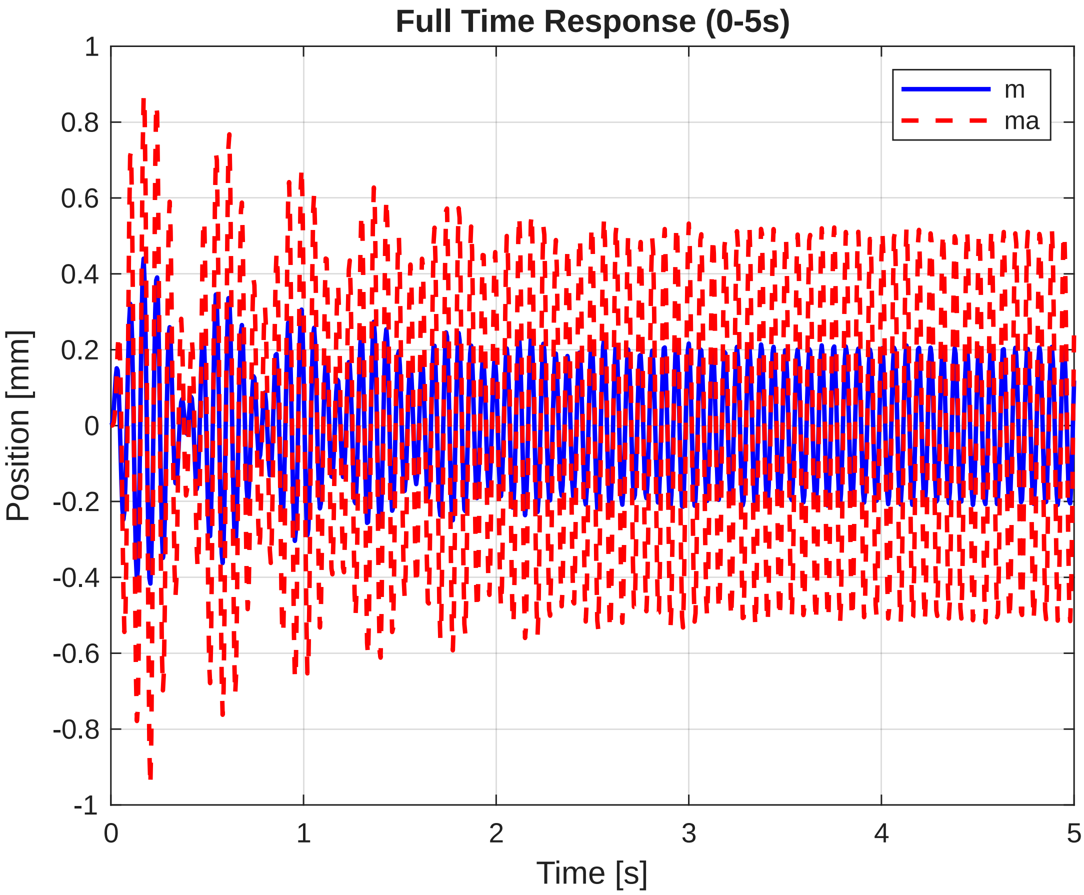

Figure 2.1: Full time response of both masses (m and ma) from 0 to 5 seconds.

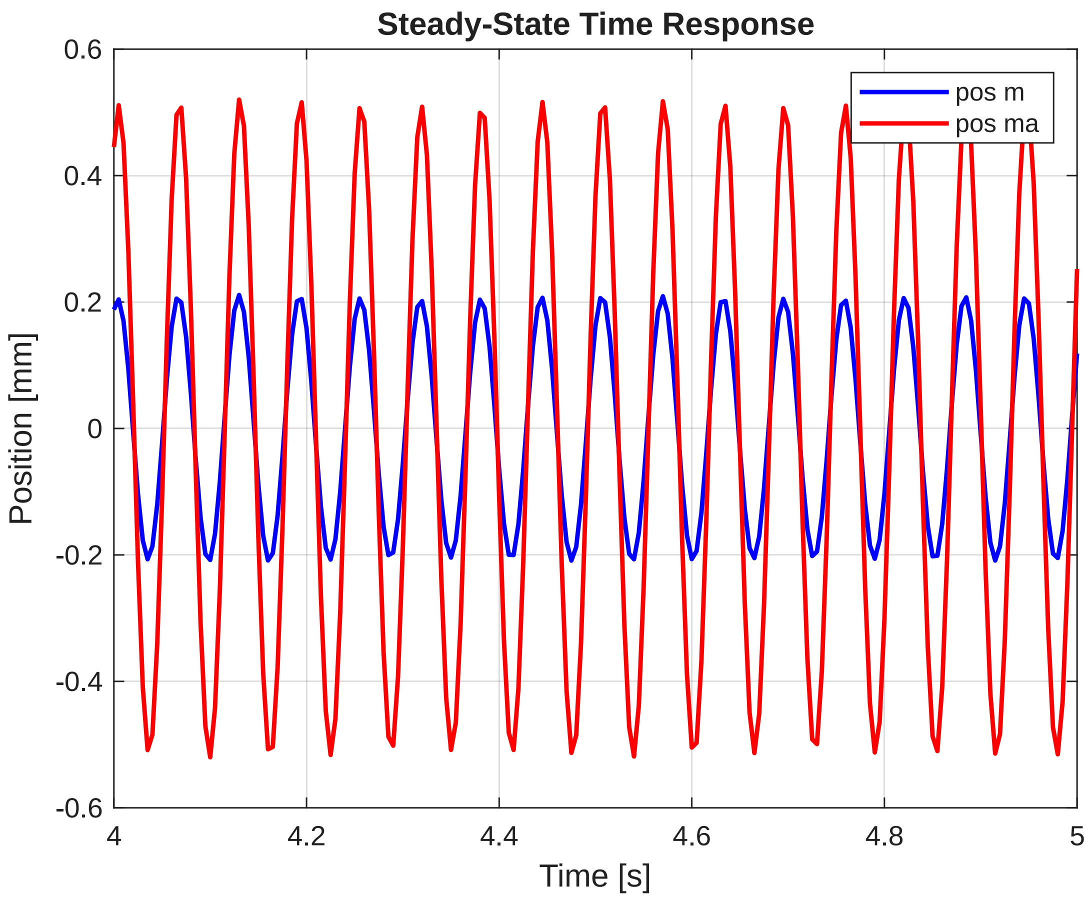

Figure 2.2: Steady-state time response of both masses.

Commentary on Figures 2.1 and 2.2

- **What it shows:** Figure 2.1 plots the displacement of both masses over 5 seconds. Figure 2.2 zooms in on the steady-state portion.
- **What it means and why:** Figure 2.1 shows two distinct phases:
  1. **Transient Response (approx. 0-2s):** The amplitudes are uneven and changing. This is the system "warming up" from rest as its natural frequencies mix with the driving frequency.
  2. **Steady-State Response (approx. 2-5s):** Both masses settle into a smooth, stable oscillation with a constant amplitude. The transient effects have damped out, and the system is now vibrating only at the 100 rad/s driving frequency. Figure 2.2 confirms this, showing a stable, in-phase oscillation.

2.3 Frequency Response Function (Original System)

A state-space model of the initial system was created to plot its Bode magnitude, representing the displacement of the primary mass (m) under a unit force excitation.

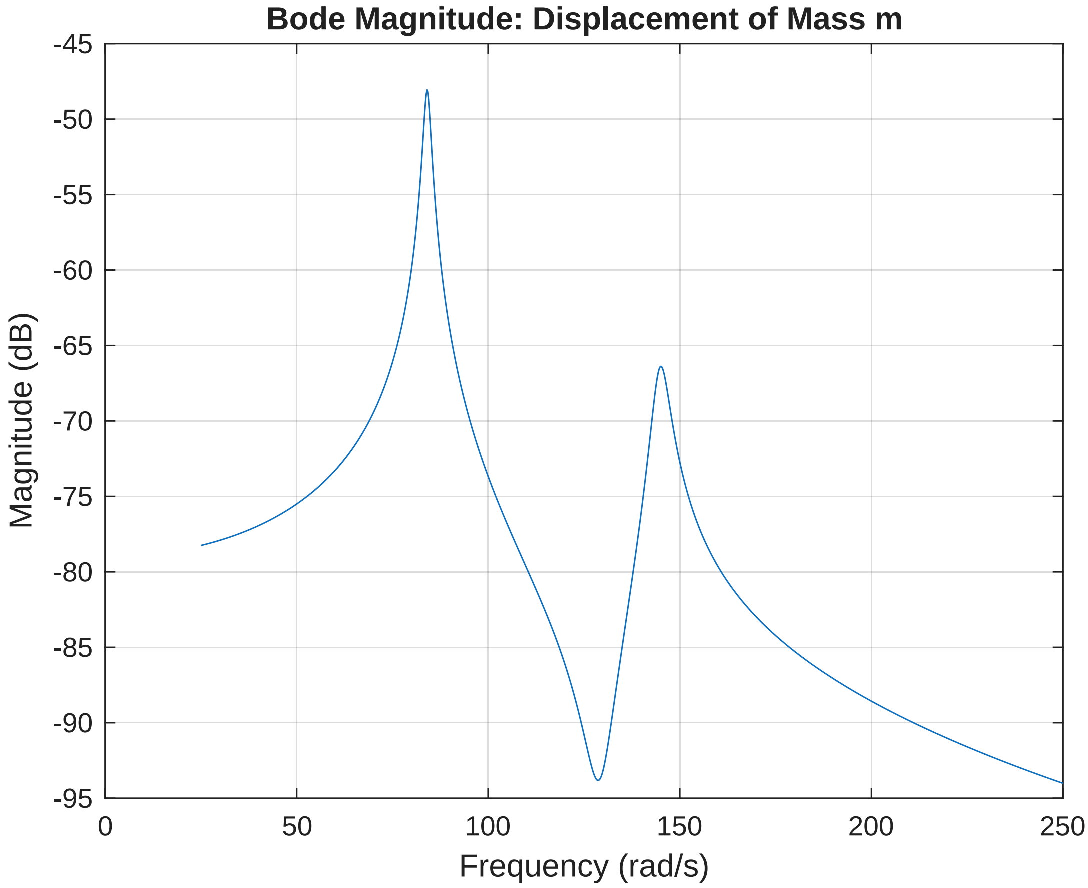

Figure 2.3: Bode magnitude diagram for the original, un-optimized system parameters.

Commentary on Figure 2.3

- **What it shows:** This is the **Frequency Response Function (FRF)**. It plots the steady-state amplitude (in dB) of the primary mass (m) for a wide range of driving frequencies (25 to 250 rad/s).

- **What it means and why:** This graph is the **key problem statement**. It shows two very tall, sharp peaks at approximately 84 rad/s and 145 rad/s. These are the **resonant frequencies** of the 2-DOF system. If the system is excited by a force at or near these two frequencies, the vibration amplitude becomes dangerously large. Our goal is to suppress these peaks.

3 Optimization of the Tuned Mass Damper (TMD)

Here, we find the optimal stiffness (k_a) and damping (c_a) for the absorber to solve the problem identified in Figure 2.3.

3.1 Sine-Sweep Response (Original System)

A sine-sweep input (25-250 rad/s) was used to excite the system over 100 seconds.

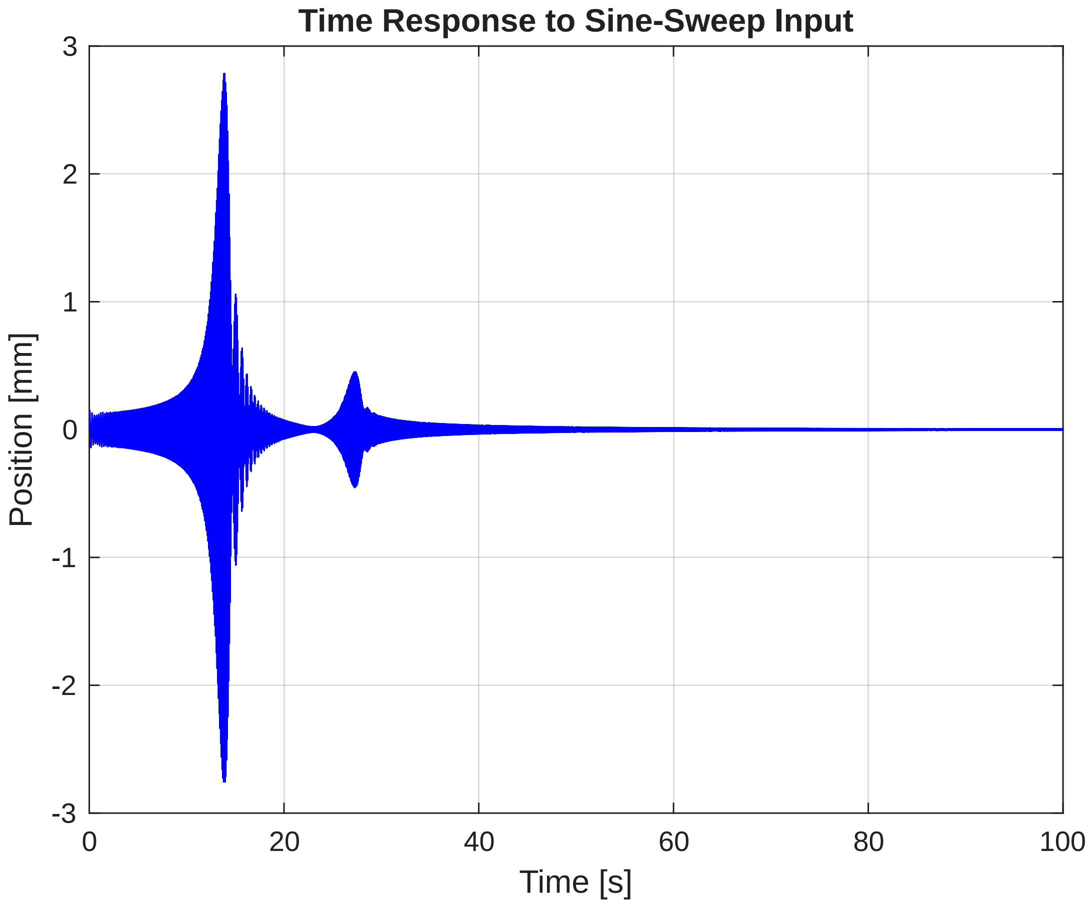

Figure 3.1: Time response of the original system to a sine-sweep input.

Commentary on Figure 3.1

- **What it shows:** This graph shows the displacement of mass m as the driving frequency is slowly swept from 25 to 250 rad/s.
- **What it means and why:** This is the time-domain equivalent of the FRF. The amplitude of the vibration (the "envelope" of the blue line) directly "traces" the FRF from Figure 2.3. You can clearly see two "bursts" of high-amplitude vibration, corresponding exactly to the moments when the sweep frequency passed through the two resonant peaks.

3.2 Numerical Optimization (H_∞ Norm)

We used `fmincon` to find the k_a and c_a that minimize the maximum peak of the FRF (H_∞ optimization).

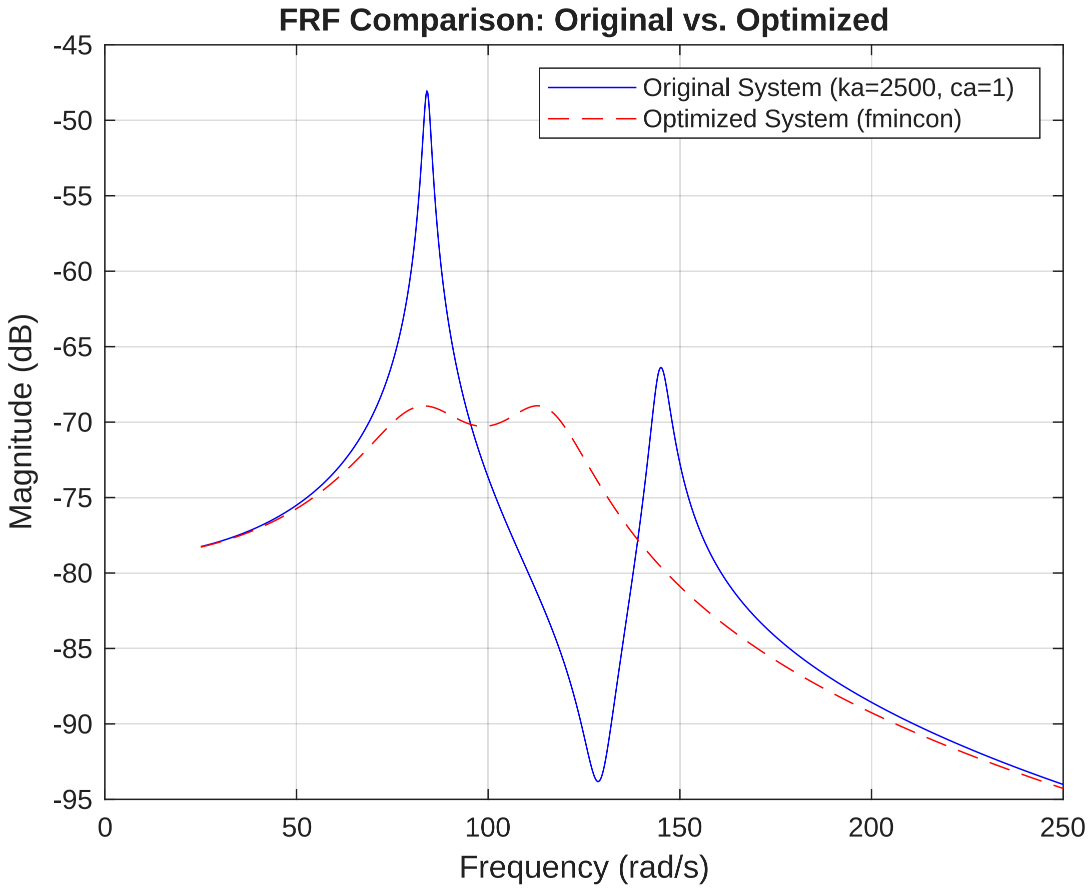

Figure 3.2: FRF Comparison: Original (blue) vs. H_∞ Optimized (red).

Commentary on Figure 3.2

- **What it shows:** This is the **most important graph of Section 2**. It overlays the original, problematic FRF (blue) with the new FRF (red dashed line) after the optimization.

- **What it means and why:** The optimization was a complete success. The two tall, sharp peaks of the original system are **gone**. They have been replaced by two new, much shorter peaks of approximately equal height. The optimization has "flattened" the response, drastically reducing the worst-case vibration amplitude.

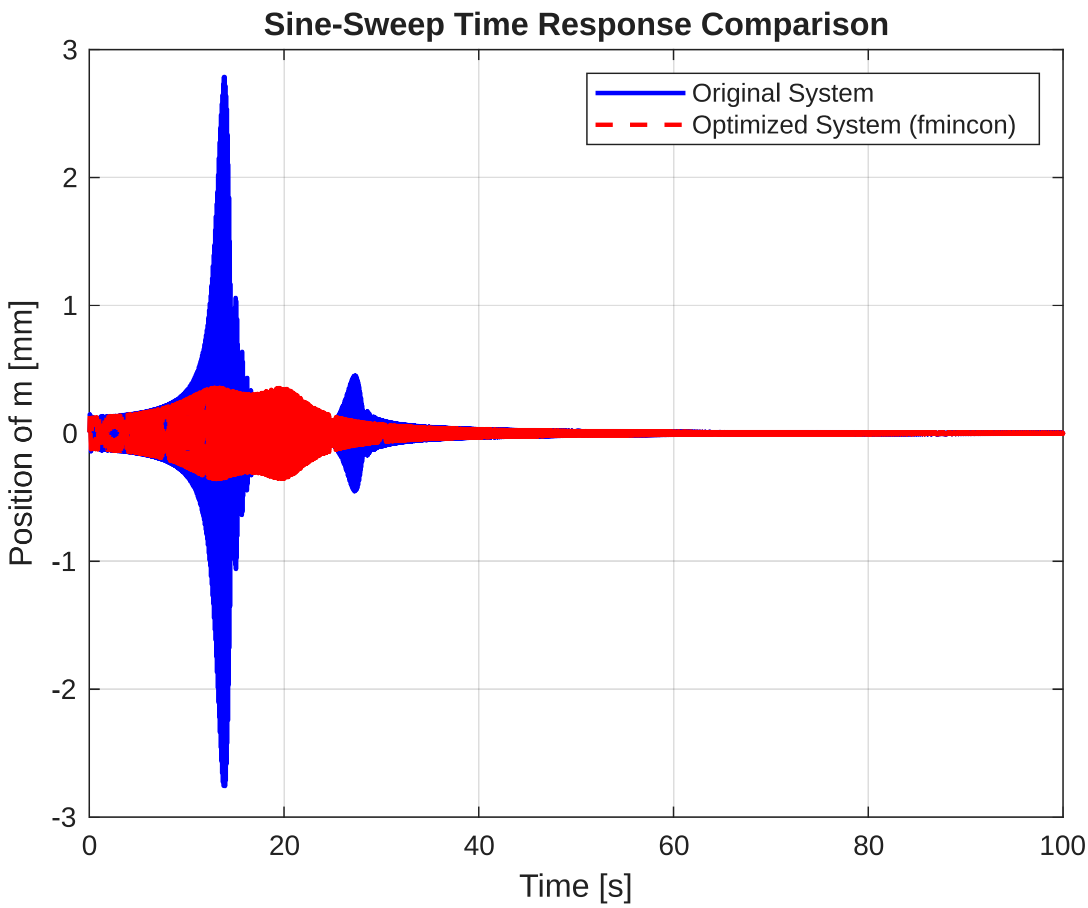

Figure 3.3: Sine-sweep time response comparison: Original vs. Optimized.

Commentary on Figure 3.3

- **What it shows:** This plot compares the sine-sweep response of the original system (blue) with the new optimized system (red dashed line).
- **What it means and why:** This visually confirms the success shown in the FRF. The two large "bursts" of vibration in the blue line are **completely suppressed** in the red line. The optimized system's response is low and controlled across the entire frequency range.

3.3 Comparison with Liu & Liu (2005) Tuning

The numerical optimization was compared to the analytical tuning formulas from Liu & Liu (2005).

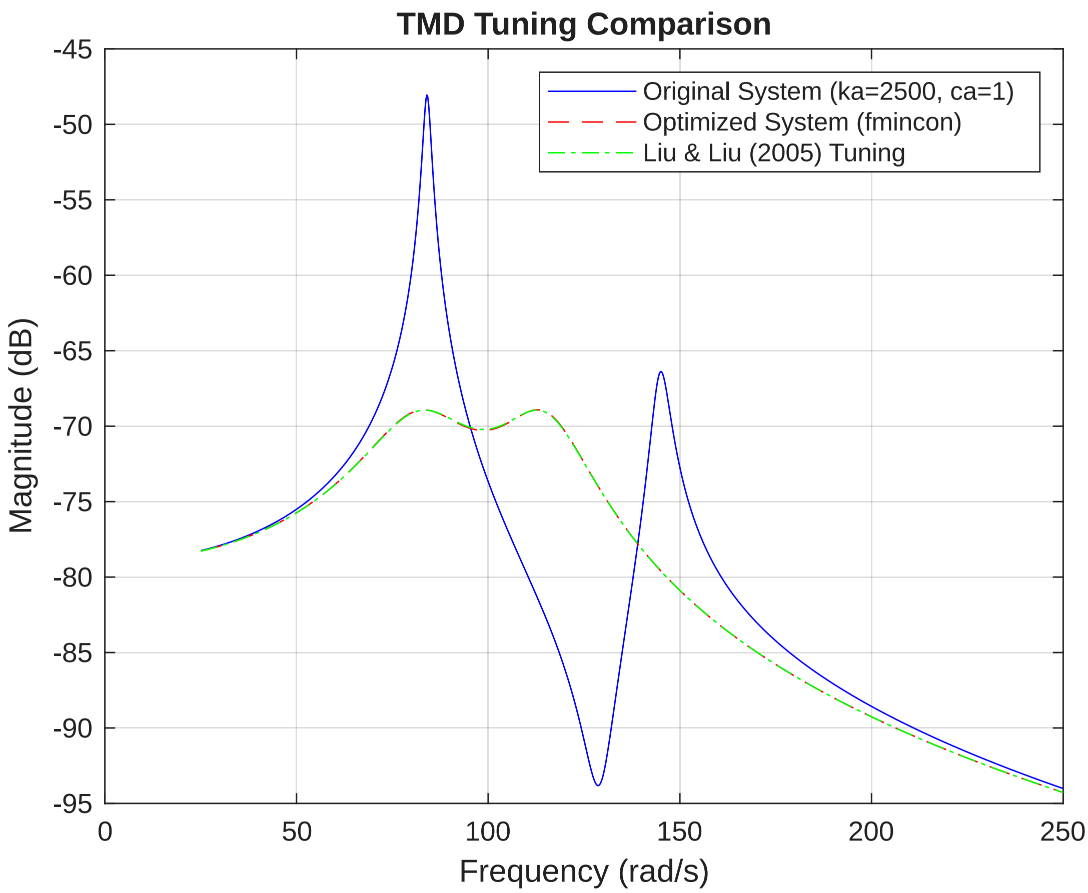

Figure 3.4: Comparison of all TMD tunings: Original, fmincon Optimized, and Liu & Liu.

Commentary on Figure 3.4

- **What it shows:** This graph overlays three FRFs: Original (blue), Optimized `fmincon` (red dashed), and the Liu & Liu analytical tuning (green dotted).
- **What it means and why:** The red and green lines are **almost perfectly identical**. This is a critical validation. It means our numerical optimization and the established analytical formula arrived at the exact same solution. This gives us high confidence that our optimization is correct and that the Liu & Liu model is accurate.

4 Equivalent Piezoelectric Resonant Shunt

This section establishes the analogy: a mechanical TMD (m_a, c_a, k_a) is equivalent to a piezoelectric R-L shunt circuit $(L, R, K_D - K_E)$.

- Absorber Mass (m_a) $\leftrightarrow$ **Inductance** (L)
- Absorber Damper (c_a) $\leftrightarrow$ **Resistance** (R)
- Absorber Spring (k_a) $\leftrightarrow$ **Piezo Stiffness** $(K_D - K_E)$

4.1 Piezoelectric Beam Modal Analysis

Experimental FRF data for a piezoelectric beam was loaded for its Short-Circuit (SC) and Open-Circuit (OC) states.

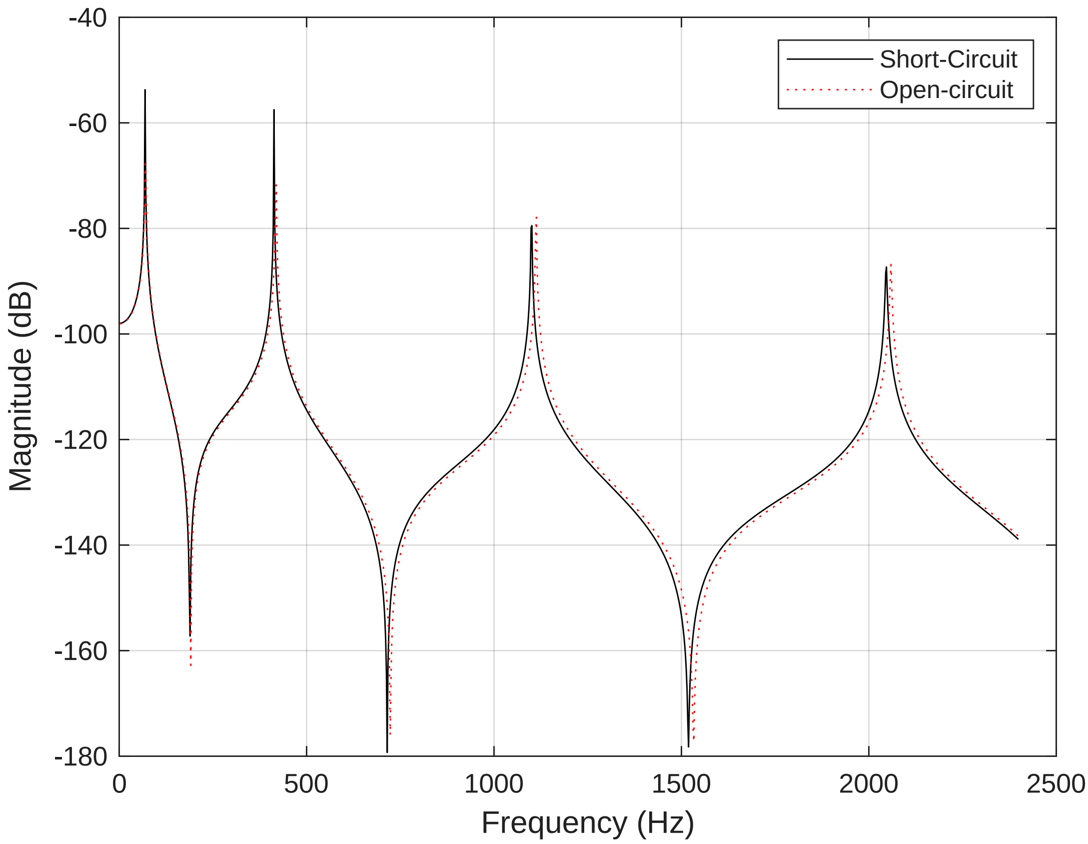

Figure 4.1: Experimental FRF data for Short-Circuit (SC) and Open-Circuit (OC) configurations.

Commentary on Figure 4.1

- **What it shows:** This is real experimental data from a piezoelectric beam, showing its FRF in two electrical states: Short-Circuit (SC, black) and Open-Circuit (OC, red).

- **What it means and why:** This graph provides the parameters for our model.
  - **Short-Circuit (SC):** Resonance at a lower frequency. This is the beam's baseline mechanical stiffness, K_E.
  - **Open-Circuit (OC):** Resonance shifts to a *higher* frequency. The piezoelectric effect adds stiffness. This higher stiffness is K_D.
- The "gap" between these two peaks ($K_D - K_E$) is the "piezoelectric stiffness," which is our k_a in the analogy.

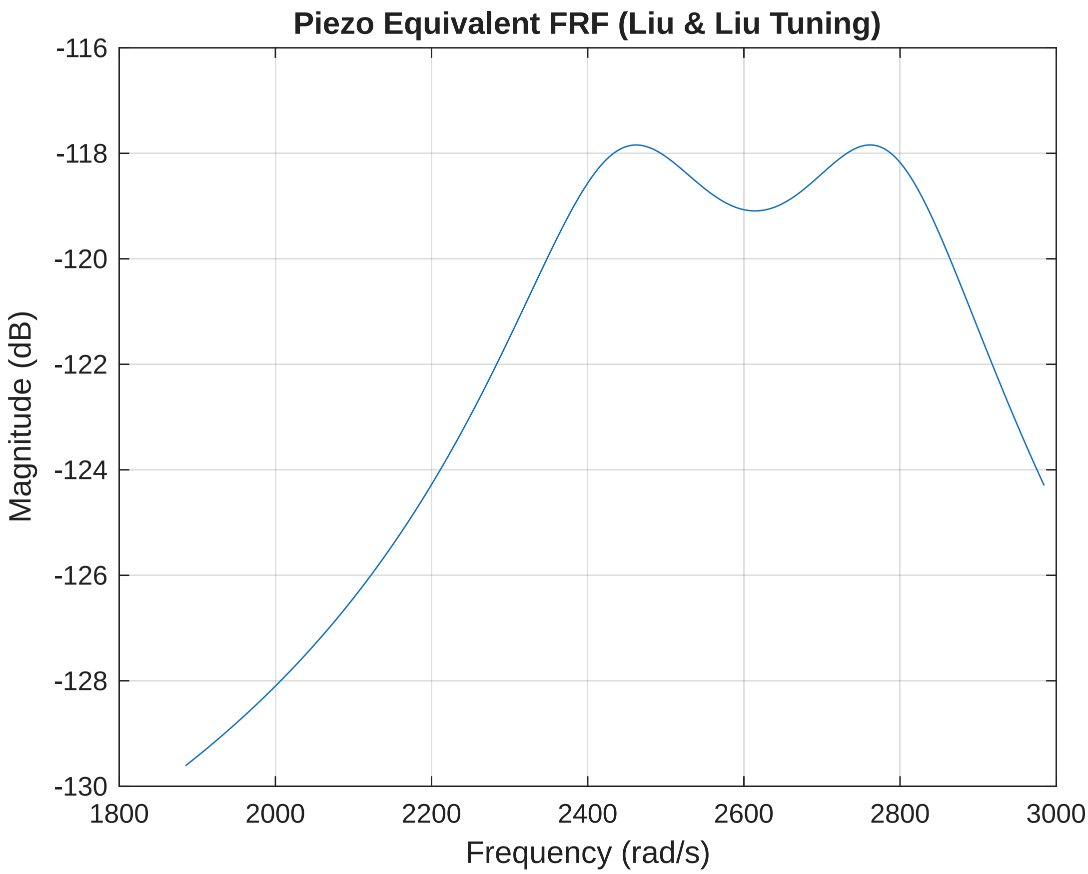

Figure 4.2: FRF of the equivalent mechanical system, tuned using Liu & Liu analogy.

Commentary on Figure 4.2

- **What it shows:** This is the FRF of our model of the piezoelectric system, tuned using the Liu & Liu mechanical rules via the electromechanical analogy.
- **What it means and why:** The plot shows the characteristic two-peak, well-damped response, just like our optimized mechanical system. This is the **proof-of-concept** that the analogy works. We can use mechanical tuning rules to design an effective electrical damping circuit.

4.2 Comparison of Shunt Tuning Rules

We now compare the tuning from the mechanical analogy (Liu & Liu) to a direct electrical tuning formula (Thomas et al.).

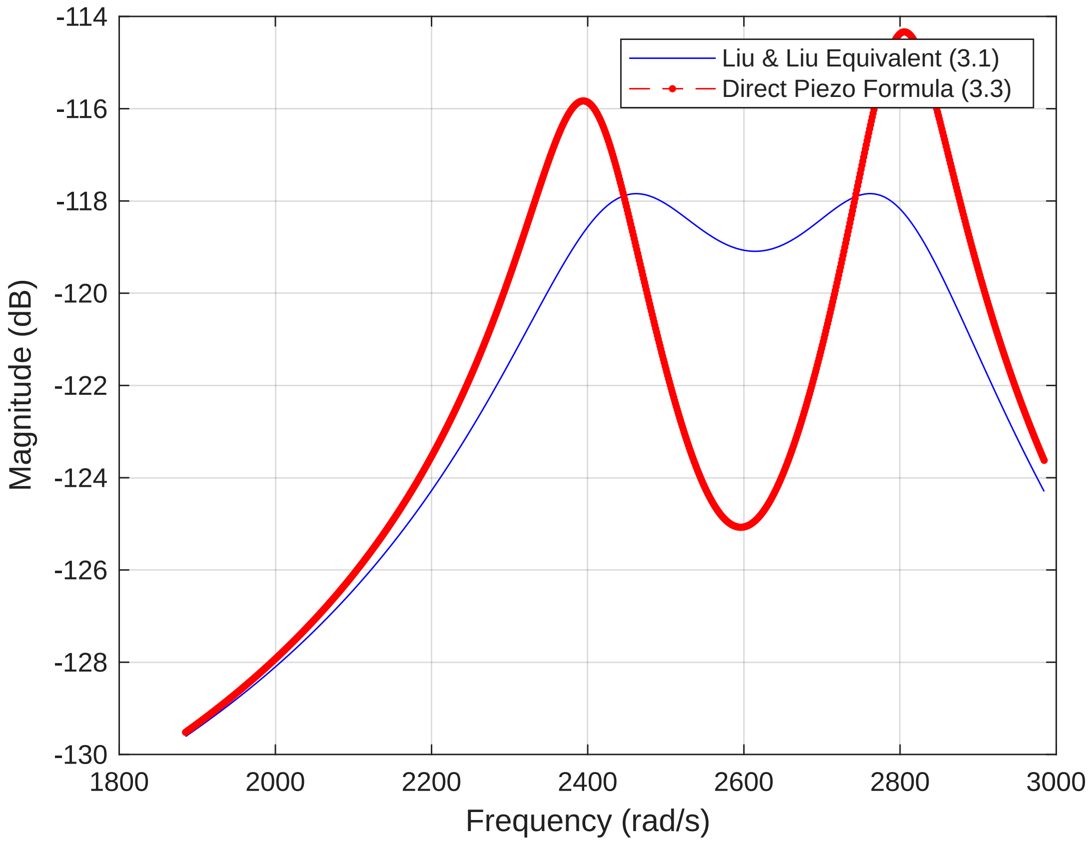

Figure 4.3: Comparison of FRFs from Liu & Liu analogy and direct piezo formulas.

Commentary on Figure 4.3

- **What it shows:** This plot compares the FRF from our Liu & Liu mechanical analogy (blue) with the FRF from a direct piezoelectric tuning formula (red).
- **What it means and why:** Both methods are highly effective and produce similar results. They both successfully replace the single, tall resonance with two short, damped peaks. The small difference just shows that the two "optimal" formulas are based on slightly different assumptions, but both are valid design approaches.

4.3 Shunt Parameter Sensitivity Analysis

This final section investigates the system's robustness. What happens if the electrical components L and R are not perfect?

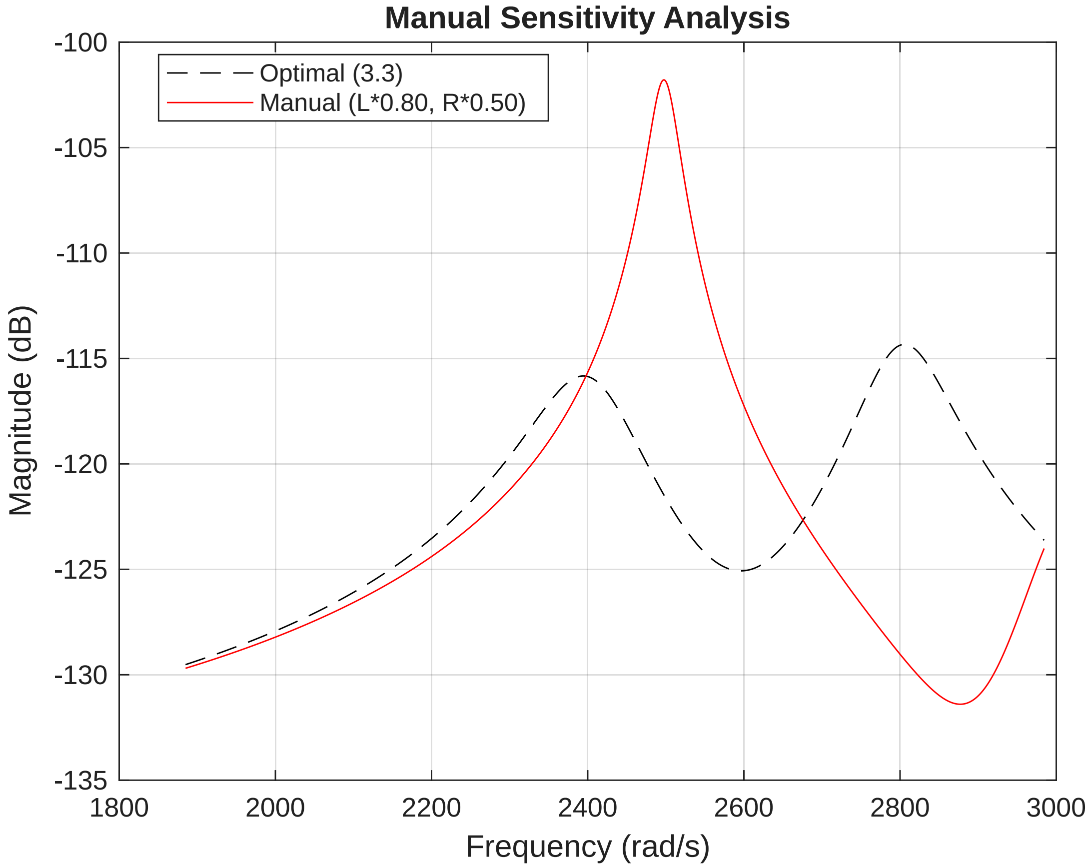

Figure 4.4: Manual sensitivity analysis (L*0.80, R*0.50).

Commentary on Figure 4.4

- **What it shows:** This compares the optimal FRF (black) with a manually detuned FRF (red), where L is 20% too low and R is 50% too low.
- **What it means and why:** This simply shows what happens when your components are "wrong." The resulting red curve is much worse (one peak is much higher), proving that the specific L and R values are critical to performance.

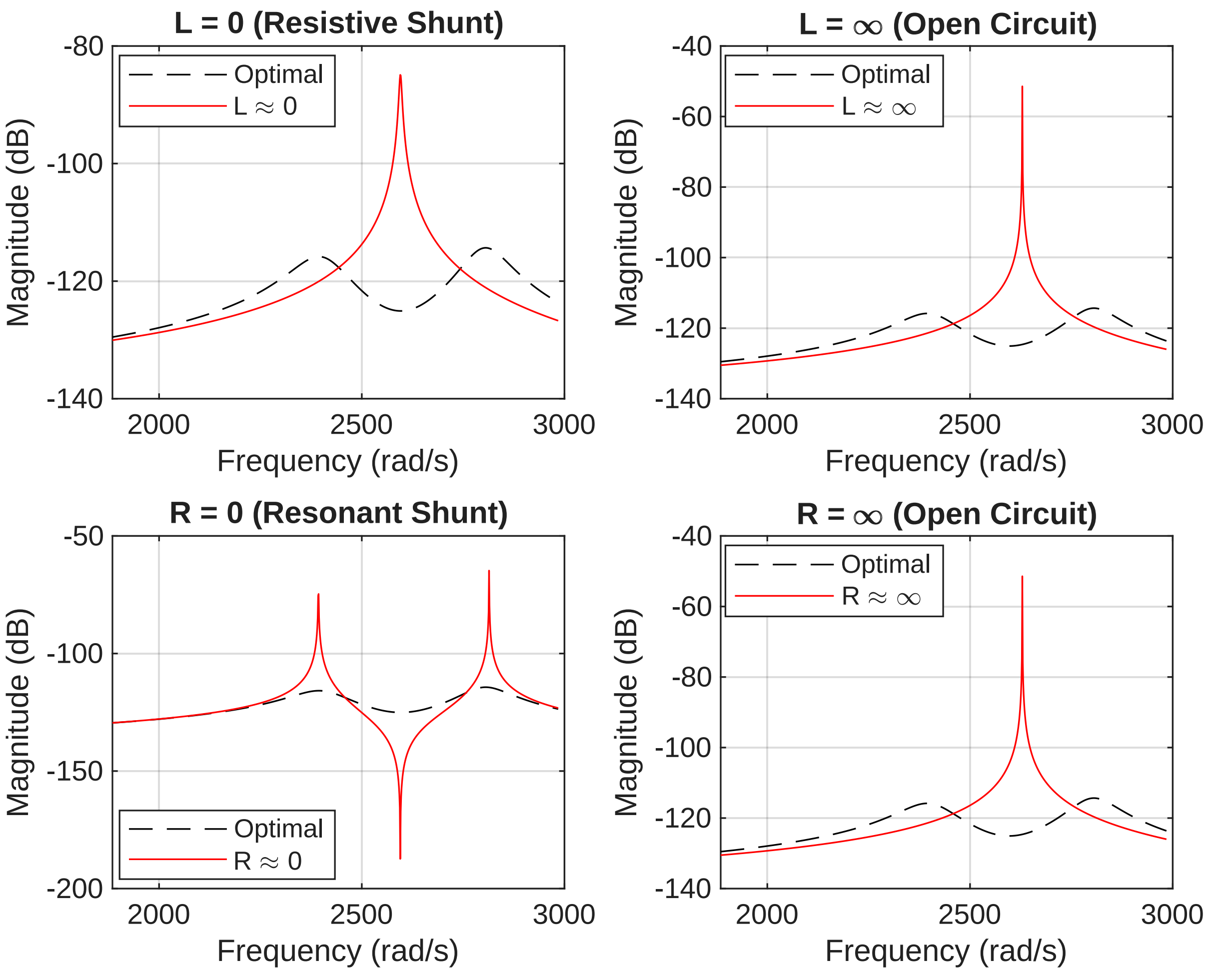

Figure 4.5: Sensitivity to extreme (null/infinite) values of L and R.

Commentary on Figure 4.5 (Extreme Cases)

This 2x2 grid is a crucial physical check of the analogy.

- **L = 0 (Resistive Shunt):** L is analogous to m_a. So $L = 0$ means $m_a = 0$. This is just a simple "R" shunt (a damper with no mass). The plot correctly shows it provides *some* damping, but it's not very effective and doesn't create the two-peak "absorber" effect.

- **R = 0 (Resonant Shunt):** R is analogous to c_a. So $R = 0$ means $c_a = 0$ (an *undamped* absorber). The plot shows it creates a perfect "anti-resonance" (notch) that *deletes* the vibration at that frequency. However, it also creates two new *undamped, infinite peaks* on either side. This just moves the problem; it doesn't solve it.

- **L = ∞ or R = ∞ (Open Circuit):** An infinite L or R means infinite impedance, so no current can flow. This is the definition of an "Open Circuit" (OC). The plot correctly shows that the system reverts to the single, high-frequency OC peak (seen in Fig 4.1).

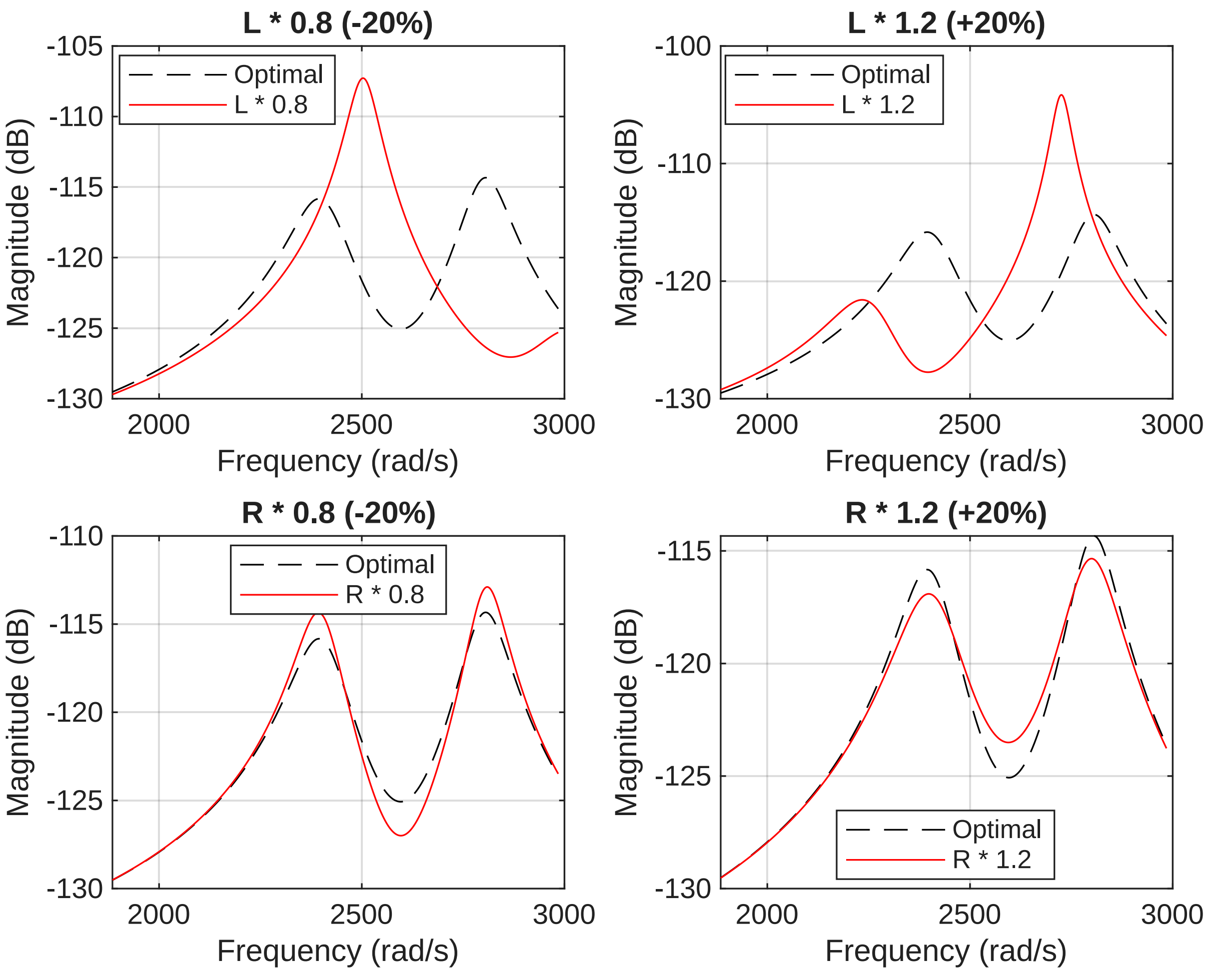

Figure 4.6: Sensitivity to ±20% variation in L and R parameters.

Commentary on Figure 4.6 (±20% Variation)

This 2x2 grid is the most important for practical design, as it shows robustness.

- **Top Row (Varying L - Inductance):** L controls the **tuning frequency** of the absorber. If L is wrong, the absorber is "mistuned." This causes the two damped peaks to become **uneven**. One peak gets higher, and the other gets lower. This ruins the "min-max" optimization.

- **Bottom Row (Varying R - Resistance):** R controls the **damping** of the absorber.

  - **R * 0.8 (Under-damped):** With not enough resistance, the energy isn't dissipated. The two peaks get *taller and sharper*.

  - **R * 1.2 (Over-damped):** With too much resistance, the absorber is "locked up" and can't move freely to absorb energy. The two peaks "smear" together, and the "valley" between them rises.

This analysis confirms the independent and critical roles of L for **tuning** (peak position) and R for **damping** (peak height).

5 Conclusion

This analysis successfully demonstrated the complete process of vibration damping, from modeling a simple mechanical system to designing and validating a piezoelectric resonant shunt.

We began by identifying the two problematic resonant frequencies of an un-optimized 2-DOF mechanical system. We then demonstrated that these resonances can be effectively suppressed using a Tuned Mass Damper (TMD). A key finding was that both a numerical H_∞ optimization (using `fmincon`) and the analytical formulas from Liu & Liu (2005) converged to an identical optimal design, validating both the optimization approach and the analytical model.

The core of this work was establishing the powerful electromechanical analogy between the mechanical TMD and a piezoelectric patch with an R-L shunt circuit. We proved that the absorber's mass (m_a) is directly analogous to the shunt's inductance (L), and the absorber's damper (c_a) is analogous to the shunt's resistance (R). This analogy was then successfully used to design an R-L shunt circuit that suppressed the resonance of a piezoelectric beam, with results comparable to direct electrical tuning formulas.

Finally, the sensitivity analysis confirmed the physical validity and robustness of the model. The analysis of extreme cases was particularly instructive:

- $R = 0$ (a purely resonant shunt) correctly created an anti-resonance notch but also two new, undamped infinite peaks.
- $L = 0$ (a purely resistive shunt) provided some damping but was far from optimal.
- Crucially, both $R = \infty$ and $L = \infty$ (representing an open circuit) correctly reverted the system to its single, high-frequency **Open-Circuit (OC)** resonance, confirming the model's physical consistency.

The $\pm 20\%$ analysis further provided a clear practical takeaway: the inductor (L) is the primary **tuning** parameter (controlling peak position), while the resistor (R) is the primary **damping** parameter (controlling peak height).

In summary, this work provides a complete, validated framework for modeling and designing piezoelectric shunt dampers, confirming their behavior as robust and effective electromechanical equivalents of classical tuned mass dampers.